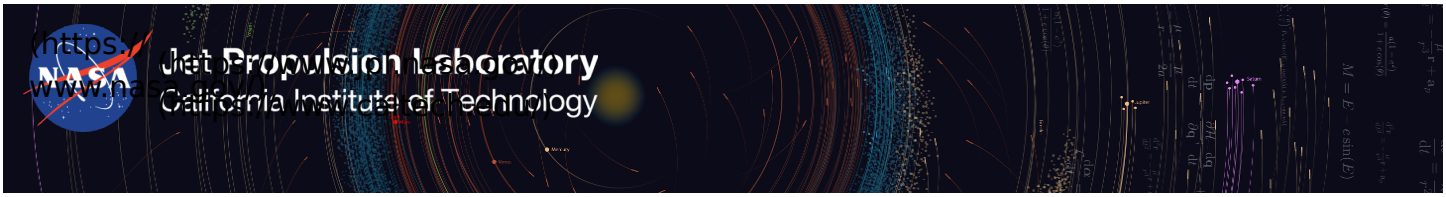




# Planetary Satellite Mean Elements

Main Table (/sats/elem/) Grouped Tables (/sats/elem/sep.html)

**Warning!** These mean orbital parameters are **not intended for ephemeris computation**. Accurate ephemerides should be obtained from our Horizons (/horizons/) system. Mean orbital parameters are primarily useful in describing the general shape and orientation of a planetary satellite's orbit.

Mean elements, in this context, are simply the elements of a precessing ellipse which has been fit in a least squares sense to the numerically integrated orbit. Please see the warning above.

Table column headings are described below.

## Satellites of Earth

mean ecliptic orbital elements
**Epoch:** 2000-01-01.5 TDB
**Ephemeris:** DE405/LE405 [ table legend ]

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) | $P_a$ (y |
|---|---|---|---|---|---|---|---|---|---|
| Moon | 301 | 384400. | 0.0554 | 318.15 | 135.27 | 5.16 | 125.08 | 27.322 | 5.9 |

## Satellites of Mars

mean orbital elements referred to the local Laplace (/glossary/lp.html) planes
**Epoch:** 2000-01-01.5 TDB
**Ephemeris:** MAR099 [ table legend ]

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) | $P_{apsis}$ (yr) |
|---|---|---|---|---|---|---|---|---|---|
| Phobos | 401 | 9375. | 0.015 | 216.3 | 189.7 | 1.1 | 169.2 | 0.3187 | 1.1 |
| Deimos | 402 | 23457. | 0.000 | 0.0 | 205.0 | 1.8 | 54.3 | 1.2625 | 0.0 |

## Satellites of Jupiter

mean orbital elements referred to the local Laplace (/glossary/lp.html) planes
**Epoch:** 2000-01-01.5 TDB
**Ephemeris:** JUP365 [ table legend ]

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) |
|---|---|---|---|---|---|---|---|---|
| Io | 501 | 421800. | 0.004 | 49.1 | 330.9 | 0.0 | 0.0 | 1.762732 |
| Europa | 502 | 671100. | 0.009 | 45.0 | 345.4 | 0.5 | 184.0 | 3.525463 |
| Ganymede | 503 | 1070400. | 0.001 | 198.3 | 324.8 | 0.2 | 58.5 | 7.155588 |
| Callisto | 504 | 1882700. | 0.007 | 43.8 | 87.4 | 0.3 | 309.1 | 16.690440 |
| Amalthea | 505 | 181400. | 0.003 | 180.1 | 310.6 | 0.4 | 282.9 | 0.499918 |
| Thebe | 514 | 221900. | 0.018 | 26.6 | 182.1 | 1.1 | 340.4 | 0.676105 |
| Adrastea | 515 | 129000. | 0.000 | 0.0 | 214.5 | 0.0 | 0.0 | 0.298260 |
| Metis | 516 | 128000. | 0.000 | 0.0 | 166.0 | 0.0 | 0.0 | 0.294779 |

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mean ecliptic orbital elements
**Epoch:** 2000-01-01.5 TDB
**Ephemeris:** JUP347 [ table legend ]

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) |
|---|---|---|---|---|---|---|---|---|
| Himalia | 506 | 11439000. | 0.160 | 321.1 | 78.3 | 28.4 | 64.2 | 249.9090 |
| Elara | 507 | 11710700. | 0.212 | 129.9 | 346.9 | 27.8 | 112.8 | 258.8861 |
| Pasiphae | 508 | 23463200. | 0.412 | 172.8 | 279.3 | 148.3 | 315.7 | 734.4215 |
| Sinope | 509 | 23679300. | 0.262 | 354.3 | 157.4 | 157.3 | 308.0 | 744.5951 |
| Lysithea | 510 | 11699100. | 0.117 | 48.3 | 328.5 | 27.7 | 9.2 | 258.5035 |

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) |
|---|---|---|---|---|---|---|---|---|
| Carme | 511 | 23139200. | 0.261 | 6.5 | 259.5 | 164.6 | 115.5 | 719.2806 |
| Ananke | 512 | 21029500. | 0.238 | 78.8 | 271.7 | 147.6 | 13.7 | 623.1097 |
| Leda | 513 | 11145200. | 0.162 | 271.3 | 232.6 | 28.2 | 216.9 | 240.3264 |
| Callirrhoe | 517 | 23789400. | 0.290 | 13.5 | 117.9 | 144.9 | 278.2 | 749.7910 |
| Themisto | 518 | 7397000. | 0.257 | 236.3 | 289.1 | 44.3 | 202.7 | 129.9681 |
| Magaclite | 519 | 23640100. | 0.421 | 309.9 | 116.1 | 149.9 | 292.2 | 742.7715 |
| Taygete | 520 | 23103400. | 0.257 | 207.7 | 116.5 | 164.7 | 300.8 | 717.5917 |
| Chaldene | 521 | 22926300. | 0.261 | 231.6 | 275.3 | 164.7 | 134.5 | 709.3625 |
| Harpalyke | 522 | 20887500. | 0.239 | 104.3 | 255.4 | 147.8 | 35.6 | 616.7833 |
| Kalyke | 523 | 23298000. | 0.261 | 240.5 | 224.9 | 164.7 | 37.7 | 726.7007 |
| Iocaste | 524 | 21062300. | 0.223 | 91.4 | 182.0 | 148.7 | 268.9 | 624.5479 |
| Erinome | 525 | 23027200. | 0.272 | 359.9 | 273.1 | 164.3 | 311.2 | 714.0542 |
| Isonoe | 526 | 22976300. | 0.249 | 131.0 | 116.3 | 164.9 | 134.1 | 711.6604 |
| Praxidike | 527 | 20931100. | 0.245 | 182.5 | 128.0 | 148.2 | 281.3 | 618.7229 |
| Autonoe | 528 | 23785200. | 0.326 | 61.2 | 134.0 | 150.7 | 264.8 | 749.6097 |
| Thyone | 529 | 20972700. | 0.235 | 105.1 | 242.5 | 147.6 | 240.8 | 620.5875 |
| Hermippe | 530 | 21103600. | 0.220 | 277.2 | 158.6 | 150.2 | 330.3 | 626.3799 |
| Aitne | 531 | 23059400. | 0.273 | 72.2 | 120.6 | 164.5 | 358.4 | 715.5396 |
| Eurydome | 532 | 22894500. | 0.287 | 223.4 | 281.8 | 148.9 | 297.7 | 707.8569 |
| Euanthe | 533 | 20822900. | 0.243 | 312.4 | 353.5 | 148.1 | 265.2 | 613.9278 |
| Euporie | 534 | 19261900. | 0.148 | 109.5 | 56.4 | 145.5 | 63.4 | 546.1778 |
| Orthosie | 535 | 20897800. | 0.294 | 244.7 | 161.1 | 144.2 | 212.4 | 617.2347 |
| Sponde | 536 | 23538700. | 0.323 | 56.6 | 177.4 | 149.4 | 109.3 | 737.9542 |
| Kale | 537 | 23047800. | 0.262 | 69.2 | 175.9 | 164.6 | 56.7 | 715.0160 |
| Pasithee | 538 | 22840800. | 0.274 | 207.9 | 237.8 | 164.5 | 318.0 | 705.4090 |
| Hegemone | 539 | 23342600. | 0.357 | 192.2 | 241.1 | 152.5 | 311.5 | 728.7743 |
| Mneme | 540 | 20815800. | 0.240 | 50.1 | 236.1 | 147.8 | 0.9 | 613.6104 |
| Aoede | 541 | 23773100. | 0.437 | 43.1 | 207.6 | 155.7 | 156.5 | 749.0708 |
| Thelxinoe | 542 | 20972300. | 0.229 | 307.0 | 275.1 | 150.7 | 174.9 | 620.5458 |
| Arche | 543 | 23093200. | 0.263 | 172.5 | 34.0 | 164.5 | 333.3 | 717.1056 |
| Kallichore | 544 | 23017100. | 0.253 | 352.6 | 66.8 | 164.7 | 23.9 | 713.5931 |
| Helike | 545 | 20911400. | 0.155 | 305.1 | 45.8 | 154.4 | 94.6 | 617.8625 |
| Carpo | 546 | 17039500. | 0.415 | 90.5 | 336.6 | 53.3 | 52.1 | 454.4000 |
| Eukelade | 547 | 23062400. | 0.274 | 290.1 | 234.2 | 164.7 | 200.4 | 715.6868 |
| Cyllene | 548 | 23650000. | 0.421 | 174.6 | 153.3 | 146.8 | 253.6 | 743.2062 |
| Kore | 549 | 24203300. | 0.338 | 135.3 | 40.6 | 141.7 | 324.2 | 769.4229 |

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) |
|---|---|---|---|---|---|---|---|---|
| Herse | 550 | 23146700. | 0.258 | 344.1 | 123.1 | 164.4 | 299.5 | 719.6264 |
| S2010_J_1 | 551 | 23185600. | 0.256 | 175.6 | 183.4 | 164.5 | 284.7 | 721.4257 |
| S2010_J_2 | 552 | 20786900. | 0.244 | 28.8 | 292.5 | 148.0 | 357.3 | 612.3507 |
| Dia | 553 | 12257900. | 0.232 | 164.1 | 320.9 | 29.1 | 293.9 | 277.2472 |
| S2016_J_1 | 554 | 20796700. | 0.245 | 298.5 | 215.3 | 145.1 | 247.7 | 612.7812 |
| S2003_J_18 | 555 | 20332800. | 0.102 | 86.2 | 252.5 | 145.7 | 166.8 | 592.3333 |
| S2011_J_2 | 556 | 22903400. | 0.358 | 262.9 | 297.6 | 151.7 | 24.9 | 708.2931 |
| Eirene | 557 | 23051300. | 0.263 | 86.4 | 336.7 | 164.7 | 180.0 | 715.1910 |
| Philophrosyn | 558 | 22600200. | 0.221 | 0.9 | 88.9 | 146.1 | 234.3 | 694.2042 |
| S2017_J_1 | 559 | 23739600. | 0.321 | 74.7 | 250.1 | 145.6 | 251.8 | 747.4382 |
| Eupheme | 560 | 20763400. | 0.234 | 73.6 | 339.8 | 147.9 | 227.5 | 611.3160 |
| S2003_J_19 | 561 | 23153100. | 0.264 | 186.8 | 196.9 | 164.6 | 21.9 | 719.9222 |
| Valetudo | 562 | 18690100. | 0.217 | 316.3 | 90.5 | 34.5 | 291.2 | 522.0743 |
| S2017_J_2 | 563 | 22949600. | 0.270 | 153.0 | 278.2 | 164.5 | 359.4 | 710.4208 |
| S2017_J_3 | 564 | 20936500. | 0.238 | 105.7 | 235.4 | 147.9 | 27.9 | 618.9653 |
| Pandia | 565 | 11479600. | 0.178 | 188.7 | 158.5 | 28.9 | 249.8 | 251.2319 |
| S2017_J_5 | 566 | 23202000. | 0.261 | 292.5 | 69.3 | 164.7 | 42.3 | 722.1972 |
| S2017_J_6 | 567 | 23251200. | 0.333 | 15.5 | 146.0 | 149.6 | 312.8 | 724.4688 |
| S2017_J_7 | 568 | 20960400. | 0.235 | 288.7 | 353.0 | 147.4 | 264.9 | 620.0167 |
| S2017_J_8 | 569 | 22819600. | 0.259 | 330.5 | 349.4 | 164.8 | 97.3 | 704.4181 |
| S2017_J_9 | 570 | 21764200. | 0.197 | 276.4 | 243.3 | 155.4 | 245.3 | 656.0479 |
| Ersa | 571 | 11399400. | 0.117 | 300.7 | 270.8 | 29.0 | 113.7 | 248.6153 |
| S2011_J_1 | 572 | 23120800. | 0.269 | 50.4 | 225.4 | 164.7 | 253.5 | 718.4153 |
| S2003_J_2 | 55501 | 20992900. | 0.225 | 170.6 | 42.3 | 150.1 | 348.4 | 621.4715 |
| S2003_J_4 | 55502 | 22922300. | 0.327 | 208.8 | 216.5 | 148.3 | 175.4 | 709.1229 |
| S2003_J_9 | 55503 | 23195100. | 0.268 | 288.4 | 359.8 | 164.7 | 46.3 | 721.8792 |
| S2003_J_10 | 55504 | 23384400. | 0.257 | 187.6 | 272.8 | 164.6 | 155.7 | 730.7375 |
| S2003_J_12 | 55505 | 20959300. | 0.235 | 66.1 | 160.7 | 150.0 | 54.1 | 619.9611 |
| S2003_J_16 | 55506 | 20877500. | 0.238 | 55.6 | 298.2 | 147.8 | 5.3 | 616.3444 |
| S2003_J_23 | 55507 | 23824000. | 0.306 | 256.6 | 140.6 | 144.4 | 30.9 | 751.3993 |
| S2003_J_24 | 55508 | 22882400. | 0.263 | 84.1 | 121.6 | 164.6 | 208.7 | 707.3347 |
| S2011_J_3 | 55509 | 11716800. | 0.192 | 134.7 | 359.4 | 27.6 | 123.3 | 259.0875 |
| S2018_J_2 | 55510 | 11419700. | 0.152 | 288.6 | 115.5 | 28.3 | 91.8 | 249.2750 |
| S2018_J_3 | 55511 | 23400200. | 0.268 | 177.5 | 279.0 | 164.9 | 135.7 | 731.4875 |
| S2021_J_1 | 55512 | 20954700. | 0.228 | 20.6 | 129.1 | 150.5 | 241.1 | 619.7687 |
| S2021_J_2 | 55513 | 20926600. | 0.242 | 25.9 | 110.8 | 148.1 | 269.5 | 618.5028 |

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) |
|---|---|---|---|---|---|---|---|---|
| S2021_J_3 | 55514 | 20776600. | 0.239 | 218.6 | 167.9 | 147.9 | 141.7 | 611.8736 |
| S2021_J_4 | 55515 | 23019700. | 0.265 | 140.5 | 218.9 | 164.6 | 129.9 | 713.7056 |
| S2021_J_5 | 55516 | 23414600. | 0.272 | 273.4 | 236.5 | 164.9 | 154.4 | 732.1528 |
| S2021_J_6 | 55517 | 22870400. | 0.271 | 146.2 | 208.5 | 164.9 | 87.2 | 706.7653 |
| S2016_J_3 | 55518 | 22719300. | 0.251 | 94.7 | 59.1 | 164.6 | 36.8 | 699.7583 |
| S2016_J_4 | 55519 | 23113900. | 0.294 | 249.4 | 217.5 | 147.1 | 203.2 | 718.0382 |
| S2018_J_4 | 55520 | 16328500. | 0.177 | 285.0 | 138.3 | 50.2 | 52.1 | 426.2646 |
| S2022_J_1 | 55521 | 22744700. | 0.257 | 232.4 | 42.2 | 164.5 | 310.9 | 700.9333 |
| S2022_J_2 | 55522 | 23073400. | 0.263 | 357.0 | 98.9 | 164.7 | 47.7 | 716.2104 |
| S2022_J_3 | 55523 | 21015100. | 0.248 | 351.6 | 271.7 | 148.1 | 233.1 | 622.4361 |
| S2017_J_10 | 55525 | 21075800. | 0.209 | 195.3 | 157.7 | 145.1 | 329.6 | 625.1493 |
| S2017_J_11 | 55526 | 22991300. | 0.268 | 25.5 | 236.9 | 164.8 | 134.9 | 712.3826 |

mean ecliptic orbital elements
**Epoch:** 2000-01-01.5 TDB
**Ephemeris:** JUP348 [ table legend ]

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) |
|---|---|---|---|---|---|---|---|---|
| S2011_J_4 | 55527 | 11104600. | 0.128 | 46.2 | 314.6 | 28.5 | 342.4 | 239.0521 |
| S2018_J_5 | 55528 | 23269900. | 0.261 | 292.4 | 82.9 | 164.9 | 189.4 | 725.3785 |
| S2024_J_1 | 55529 | 23462100. | 0.273 | 108.8 | 147.7 | 164.7 | 272.4 | 734.3764 |
| S2011_J_5 | 55530 | 23527800. | 0.251 | 90.5 | 66.6 | 164.6 | 43.7 | 737.4646 |

## Satellites of Saturn

mean orbital elements referred to the local Laplace (/glossary/lp.html) planes
**Epoch:** 2000-01-01.5 TDB
**Ephemeris:** SAT441 [ table legend ]

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) |
|---|---|---|---|---|---|---|---|---|
| Mimas | 601 | 186000. | 0.020 | 160.4 | 275.3 | 1.6 | 66.2 | 0.942422 |
| Enceladus | 602 | 238400. | 0.005 | 119.5 | 57.0 | 0.0 | 0.0 | 1.370218 |
| Tethys | 603 | 295000. | 0.001 | 335.3 | 0.0 | 1.1 | 273.0 | 1.887802 |
| Dione | 604 | 377700. | 0.002 | 116.0 | 212.0 | 0.0 | 0.0 | 2.736916 |
| Rhea | 605 | 527200. | 0.001 | 44.3 | 31.5 | 0.3 | 133.7 | 4.517503 |

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) |
|---|---|---|---|---|---|---|---|---|
| Titan | 606 | 1221900. | 0.029 | 78.3 | 11.7 | 0.3 | 78.6 | 15.945448 |
| Hyperion | 607 | 1481500. | 0.105 | 214.0 | 122.9 | 0.6 | 87.1 | 21.276658 |
| Iapetus | 608 | 3561700. | 0.028 | 254.5 | 74.8 | 7.6 | 86.5 | 79.331002 |
| Phoebe | 609 | 12929400. | 0.164 | 240.3 | 308.0 | 175.2 | 192.7 | 550.30391( |
| Helene | 612 | 377600. | 0.007 | 194.3 | -64.5 | 0.2 | 260.8 | 2.736916 |
| Telesto | 613 | 295000. | 0.001 | 66.2 | 360.0 | 1.2 | 242.1 | 1.887802 |
| Calypso | 614 | 295000. | 0.001 | 221.6 | -0.0 | 1.5 | 326.6 | 1.887803 |
| Methone | 632 | 194700. | 0.002 | 315.5 | 0.0 | 0.0 | 0.0 | 1.009549 |
| Polydeuces | 634 | 377600. | 0.019 | 94.1 | 359.0 | 0.2 | 167.5 | 2.736916 |

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mean orbital elements referred to the local Laplace (/glossary/lp.html) planes
**Epoch:** 2000-01-01.5 TDB
**Ephemeris:** SAT415 [ table legend ]

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) | P ( |
|---|---|---|---|---|---|---|---|---|---|
| Janus | 610 | 151500. | 0.007 | 11.1 | 111.7 | 0.2 | 159.9 | 0.697353 | 0 |
| Epimetheus | 611 | 151400. | 0.020 | 96.3 | 197.2 | 0.3 | 189.8 | 0.697012 | 0 |
| Atlas | 615 | 137700. | 0.001 | 82.3 | 289.1 | 0.0 | 0.0 | 0.604602 | 0 |
| Prometheus | 616 | 139400. | 0.002 | 341.9 | 135.4 | 0.0 | 0.0 | 0.615878 | 0 |
| Pandora | 617 | 141700. | 0.004 | 217.9 | 123.9 | 0.0 | 0.0 | 0.631369 | 0 |
| Pan | 618 | 133600. | 0.000 | 0.0 | 146.6 | 0.0 | 0.0 | 0.575051 | 0 |
| Pallene | 633 | 212300. | 0.004 | 194.9 | 2.0 | 0.2 | 304.3 | 1.156059 | 0 |
| Daphnis | 635 | 136500. | 0.000 | 0.0 | 153.6 | 0.0 | 0.0 | 0.594080 | 0 |
| Anthe | 649 | 198100. | 0.002 | 293.4 | 351.4 | 0.0 | 0.0 | 1.038898 | 1 |
| Aegaeon | 653 | 167500. | 0.000 | 0.0 | 4.4 | 0.0 | 0.0 | 0.808115 | 0 |

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mean ecliptic orbital elements
**Epoch:** 2000-01-01.5 TDB
**Ephemeris:** SAT456 [ table legend ]

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) |
|---|---|---|---|---|---|---|---|---|
| Ymir | 619 | 22955600. | 0.338 | 45.3 | 167.5 | 172.3 | 204.5 | 1298.6819 |
| Paaliaq | 620 | 14997700. | 0.378 | 237.6 | 318.2 | 48.5 | 353.2 | 685.7153 |
| Tarvos | 621 | 18216600. | 0.522 | 282.9 | 64.5 | 37.8 | 95.7 | 917.9840 |

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) |
|---|---|---|---|---|---|---|---|---|
| Ijiraq | 622 | 11344700. | 0.293 | 68.6 | 83.4 | 49.2 | 152.5 | 451.1201 |
| Suttungr | 623 | 19391900. | 0.116 | 65.0 | 171.0 | 175.7 | 253.2 | 1008.2451 |
| Kiviuq | 624 | 11307400. | 0.275 | 91.8 | 183.2 | 48.0 | 352.7 | 448.9076 |
| Mundilfari | 625 | 18588200. | 0.211 | 305.4 | 299.0 | 167.1 | 79.2 | 946.2903 |
| Albiorix | 626 | 16329200. | 0.482 | 62.8 | 155.0 | 36.8 | 111.1 | 779.0701 |
| Skathi | 627 | 15575400. | 0.281 | 202.3 | 269.3 | 151.6 | 284.1 | 725.7313 |
| Erriapus | 628 | 17506900. | 0.475 | 279.8 | 14.3 | 37.1 | 146.2 | 864.9236 |
| Siarnaq | 629 | 17881100. | 0.308 | 68.4 | 62.7 | 47.8 | 64.0 | 892.6799 |
| Thrymr | 630 | 20330500. | 0.467 | 92.2 | 216.2 | 175.0 | 247.3 | 1082.2292 |
| Narvi | 631 | 19285000. | 0.441 | 177.9 | 111.5 | 142.2 | 179.0 | 999.9417 |
| Aegir | 636 | 20664400. | 0.255 | 244.7 | 23.5 | 166.1 | 181.9 | 1109.1278 |
| Bebhionn | 637 | 17027300. | 0.459 | 5.1 | 128.4 | 38.6 | 197.7 | 829.6437 |
| Bergelmir | 638 | 19268100. | 0.145 | 133.9 | 255.7 | 158.8 | 207.9 | 998.6187 |
| Bestla | 639 | 20337800. | 0.486 | 85.6 | 33.1 | 138.3 | 283.0 | 1082.9444 |
| Farbauti | 640 | 20290700. | 0.249 | 343.6 | 9.0 | 156.2 | 138.0 | 1079.1243 |
| Fenrir | 641 | 22330800. | 0.137 | 114.5 | 38.6 | 164.5 | 234.6 | 1245.9215 |
| Fornjot | 642 | 24936800. | 0.213 | 321.9 | 53.5 | 170.0 | 269.3 | 1470.3618 |
| Hati | 643 | 19695000. | 0.372 | 10.4 | 270.6 | 165.4 | 316.3 | 1032.0236 |
| Hyrrokkin | 644 | 18340900. | 0.336 | 264.8 | 214.8 | 149.9 | 40.2 | 927.4569 |
| Kari | 645 | 22032100. | 0.469 | 160.2 | 83.1 | 153.0 | 288.3 | 1220.9757 |
| Loge | 646 | 22919200. | 0.191 | 20.5 | 32.1 | 168.1 | 333.0 | 1295.5243 |
| Skoll | 647 | 17623400. | 0.463 | 186.1 | 181.2 | 159.4 | 294.0 | 873.5736 |
| Surtur | 648 | 22748000. | 0.448 | 314.4 | 9.9 | 168.4 | 255.3 | 1281.1396 |
| Jarnsaxa | 650 | 19273500. | 0.218 | 229.2 | 175.5 | 163.0 | 9.2 | 999.1271 |
| Greip | 651 | 18380000. | 0.317 | 146.2 | 355.9 | 174.2 | 335.8 | 930.4389 |
| Tarqeq | 652 | 17751000. | 0.144 | 72.2 | 292.8 | 48.7 | 93.3 | 882.8514 |
| Gridr | 654 | 19250600. | 0.187 | 271.4 | 197.0 | 163.9 | 330.8 | 997.3319 |
| Angrboda | 655 | 20591500. | 0.216 | 104.2 | 263.0 | 177.7 | 279.1 | 1103.1986 |
| Skrymir | 656 | 21447400. | 0.437 | 68.1 | 68.9 | 175.6 | 177.7 | 1172.7215 |
| Gerd | 657 | 20947500. | 0.517 | 307.2 | 49.5 | 174.4 | 260.4 | 1131.9062 |
| S2004_S_26 | 658 | 26097500. | 0.147 | 148.4 | 193.6 | 172.9 | 328.8 | 1574.2535 |
| Eggther | 659 | 19844600. | 0.157 | 130.1 | 111.7 | 165.0 | 84.5 | 1043.8035 |
| S2004_S_29 | 660 | 17063900. | 0.485 | 309.5 | 244.1 | 38.6 | 157.9 | 832.2736 |
| Beli | 661 | 20703700. | 0.087 | 263.5 | 191.7 | 158.9 | 258.7 | 1112.2687 |
| Gunnlod | 662 | 21141800. | 0.251 | 24.7 | 311.4 | 160.4 | 293.3 | 1147.7354 |
| Thiazzi | 663 | 23577500. | 0.511 | 303.2 | 334.7 | 158.8 | 73.1 | 1351.8319 |

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) |
|---|---|---|---|---|---|---|---|---|
| S2004_S_34 | 664 | 24145800. | 0.279 | 354.5 | 160.0 | 168.3 | 292.6 | 1400.9299 |
| Alvaldi | 665 | 21993800. | 0.238 | 188.1 | 38.3 | 177.4 | 321.8 | 1217.8042 |
| Geirrod | 666 | 22259400. | 0.539 | 349.6 | 202.2 | 154.4 | 127.9 | 1240.0465 |

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mean ecliptic orbital elements
**Epoch:** 2000-01-01.5 TDB
**Ephemeris:** SAT457 [ table legend ]

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) |
|---|---|---|---|---|---|---|---|---|
| S2004_S_31 | 65067 | 17497100. | 0.159 | 253.9 | 261.8 | 48.0 | 82.5 | 863.9229 |
| S2004_S_24 | 65070 | 23338200. | 0.071 | 338.4 | 122.1 | 37.4 | 354.0 | 1331.322 |
| S2004_S_28 | 65077 | 21865900. | 0.159 | 191.9 | 95.8 | 167.9 | 128.8 | 1207.183 |
| S2004_S_21 | 65079 | 23160900. | 0.394 | 174.9 | 198.7 | 153.2 | 112.3 | 1316.116 |
| S2004_S_36 | 65081 | 23390800. | 0.625 | 213.4 | 132.0 | 153.3 | 212.7 | 1335.804 |
| S2004_S_37 | 65082 | 15956300. | 0.448 | 72.4 | 187.6 | 158.2 | 135.3 | 752.5465 |
| S2004_S_39 | 65084 | 23192400. | 0.100 | 223.3 | 83.8 | 165.9 | 168.9 | 1318.741 |
| S2004_S_7 | 65085 | 21327600. | 0.511 | 54.9 | 116.9 | 164.8 | 323.8 | 1162.926 |
| S2004_S_12 | 65086 | 19801000. | 0.337 | 88.1 | 10.9 | 164.7 | 306.0 | 1040.389 |
| S2004_S_13 | 65087 | 18453700. | 0.265 | 355.4 | 51.9 | 169.0 | 217.8 | 936.0889 |
| S2004_S_17 | 65088 | 19699000. | 0.162 | 184.5 | 227.6 | 167.9 | 12.5 | 1032.369 |
| S2006_S_1 | 65089 | 18746300. | 0.105 | 159.7 | 86.3 | 156.1 | 333.6 | 958.3236 |
| S2006_S_3 | 65090 | 21353100. | 0.432 | 191.6 | 125.3 | 156.1 | 213.3 | 1165.019 |
| S2007_S_2 | 65091 | 15939100. | 0.232 | 46.4 | 10.0 | 174.0 | 107.3 | 751.3306 |
| S2007_S_3 | 65092 | 19614400. | 0.150 | 323.1 | 304.5 | 173.8 | 99.3 | 1025.698 |
| S2019_S_1 | 65093 | 11245400. | 0.383 | 86.6 | 36.5 | 49.5 | 185.6 | 445.1701 |
| S2019_S_2 | 65094 | 16560300. | 0.279 | 103.2 | 185.1 | 173.3 | 95.3 | 795.6722 |
| S2019_S_3 | 65095 | 17077400. | 0.248 | 129.2 | 150.3 | 166.9 | 243.9 | 833.1917 |
| S2020_S_1 | 65096 | 11338600. | 0.337 | 73.2 | 2.1 | 48.2 | 161.7 | 450.7722 |
| S2020_S_2 | 65097 | 17869000. | 0.152 | 130.7 | 19.2 | 170.7 | 96.2 | 891.8576 |
| S2004_S_40 | 65098 | 16075600. | 0.297 | 127.5 | 291.4 | 169.2 | 37.9 | 760.9993 |
| S2006_S_9 | 65100 | 14406700. | 0.249 | 341.9 | 65.2 | 173.0 | 36.0 | 645.5771 |
| S2007_S_5 | 65101 | 15835600. | 0.104 | 44.4 | 23.2 | 158.4 | 128.9 | 744.0056 |
| S2020_S_3 | 65102 | 18056800. | 0.142 | 317.6 | 135.8 | 46.0 | 216.8 | 905.8410 |
| S2019_S_4 | 65103 | 17951900. | 0.408 | 247.0 | 217.8 | 170.1 | 301.4 | 898.0889 |
| S2004_S_41 | 65104 | 18095400. | 0.301 | 232.6 | 343.6 | 165.7 | 43.6 | 908.8924 |

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) |
|---|---|---|---|---|---|---|---|---|
| S2020_S_4 | 65105 | 18236400. | 0.496 | 157.7 | 178.7 | 40.1 | 67.8 | 919.5174 |
| S2020_S_5 | 65106 | 18391000. | 0.220 | 285.7 | 232.0 | 48.2 | 214.9 | 931.1889 |
| S2007_S_6 | 65107 | 18545000. | 0.168 | 19.2 | 347.8 | 166.5 | 324.1 | 942.9785 |
| S2004_S_42 | 65108 | 18240700. | 0.158 | 107.2 | 325.2 | 165.7 | 3.9 | 919.8847 |
| S2006_S_10 | 65109 | 18979900. | 0.151 | 79.7 | 173.5 | 161.6 | 350.8 | 976.3438 |
| S2019_S_5 | 65110 | 19090100. | 0.216 | 321.5 | 59.4 | 158.8 | 271.1 | 984.8722 |
| S2004_S_43 | 65111 | 18935700. | 0.432 | 234.0 | 14.5 | 171.1 | 191.9 | 972.8528 |
| S2004_S_44 | 65112 | 19515400. | 0.129 | 176.4 | 292.8 | 167.7 | 39.5 | 1017.914 |
| S2004_S_45 | 65113 | 19693700. | 0.551 | 350.6 | 75.7 | 154.0 | 306.0 | 1031.860 |
| S2006_S_11 | 65114 | 19711900. | 0.143 | 159.9 | 353.1 | 174.1 | 254.4 | 1033.352 |
| S2006_S_12 | 65115 | 19570300. | 0.542 | 78.1 | 249.6 | 38.6 | 155.1 | 1022.286 |
| S2019_S_6 | 65116 | 18205600. | 0.120 | 286.9 | 185.7 | 46.4 | 154.2 | 917.1056 |
| S2006_S_13 | 65117 | 19953300. | 0.313 | 254.7 | 248.5 | 162.0 | 44.4 | 1052.316 |
| S2019_S_7 | 65118 | 20185100. | 0.233 | 6.3 | 336.0 | 174.2 | 226.6 | 1070.800 |
| S2019_S_8 | 65119 | 20287400. | 0.311 | 144.6 | 31.8 | 172.8 | 168.6 | 1078.860 |
| S2019_S_9 | 65120 | 20359700. | 0.433 | 316.4 | 244.7 | 159.5 | 114.2 | 1084.619 |
| S2004_S_46 | 65121 | 20513800. | 0.249 | 298.1 | 209.2 | 177.2 | 281.1 | 1096.988 |
| S2019_S_10 | 65122 | 20700300. | 0.248 | 352.9 | 149.5 | 163.9 | 274.7 | 1111.992 |
| S2004_S_47 | 65123 | 16050700. | 0.291 | 295.4 | 25.2 | 160.9 | 294.8 | 759.2201 |
| S2019_S_11 | 65124 | 20664200. | 0.513 | 28.2 | 190.5 | 144.6 | 327.6 | 1109.109 |
| S2006_S_14 | 65125 | 21062300. | 0.060 | 136.1 | 264.7 | 166.7 | 274.4 | 1141.270 |
| S2019_S_12 | 65126 | 20895000. | 0.476 | 240.2 | 221.1 | 167.1 | 324.7 | 1127.599 |
| S2020_S_6 | 65127 | 21253300. | 0.480 | 27.9 | 271.2 | 166.9 | 7.0 | 1156.811 |
| S2019_S_13 | 65128 | 20964500. | 0.318 | 219.8 | 138.5 | 177.3 | 302.5 | 1133.265 |
| S2005_S_4 | 65129 | 11324500. | 0.315 | 107.1 | 247.1 | 48.0 | 169.4 | 449.9250 |
| S2007_S_7 | 65130 | 15931600. | 0.217 | 149.6 | 63.4 | 169.3 | 3.8 | 750.7958 |
| S2007_S_8 | 65131 | 17048900. | 0.490 | 258.7 | 6.3 | 36.2 | 202.2 | 831.2111 |
| S2020_S_7 | 65132 | 17394000. | 0.500 | 298.8 | 215.8 | 161.4 | 175.7 | 856.5340 |
| S2019_S_14 | 65133 | 17852800. | 0.172 | 303.2 | 240.9 | 46.2 | 149.6 | 890.5868 |
| S2019_S_15 | 65134 | 21191100. | 0.257 | 17.9 | 302.9 | 157.8 | 22.0 | 1151.661 |
| S2005_S_5 | 65135 | 21364900. | 0.588 | 284.6 | 9.9 | 169.5 | 84.7 | 1165.956 |
| S2006_S_15 | 65136 | 21799600. | 0.117 | 224.8 | 313.6 | 161.1 | 12.2 | 1201.692 |
| S2006_S_16 | 65137 | 21721200. | 0.204 | 282.9 | 62.5 | 164.1 | 165.4 | 1195.128 |
| S2006_S_17 | 65138 | 22384200. | 0.425 | 149.4 | 206.6 | 168.7 | 274.1 | 1250.463 |
| S2004_S_48 | 65139 | 22137400. | 0.374 | 286.6 | 195.0 | 161.9 | 194.7 | 1229.863 |
| S2020_S_8 | 65140 | 21967200. | 0.252 | 45.0 | 181.2 | 161.8 | 41.8 | 1215.606 |

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) |
|---|---|---|---|---|---|---|---|---|
| S2004_S_49 | 65141 | 22399400. | 0.453 | 199.0 | 19.2 | 159.8 | 221.2 | 1251.684 |
| S2004_S_50 | 65142 | 22345000. | 0.450 | 197.8 | 71.5 | 164.0 | 175.9 | 1247.191 |
| S2006_S_18 | 65143 | 22760600. | 0.131 | 62.5 | 14.5 | 169.5 | 47.6 | 1282.091 |
| S2019_S_16 | 65144 | 23265200. | 0.250 | 280.6 | 351.8 | 162.0 | 206.7 | 1324.945 |
| S2019_S_17 | 65145 | 22722700. | 0.546 | 317.0 | 40.5 | 155.5 | 125.3 | 1278.961 |
| S2019_S_18 | 65146 | 23139500. | 0.509 | 304.5 | 340.4 | 154.6 | 230.6 | 1314.267 |
| S2019_S_19 | 65147 | 23044400. | 0.458 | 305.8 | 335.8 | 151.8 | 248.3 | 1306.160 |
| S2019_S_20 | 65148 | 23677900. | 0.354 | 198.6 | 281.1 | 156.0 | 176.6 | 1360.467 |
| S2006_S_19 | 65149 | 23800500. | 0.467 | 19.2 | 195.9 | 175.5 | 281.1 | 1371.020 |
| S2004_S_51 | 65150 | 25207100. | 0.201 | 73.4 | 85.1 | 171.2 | 293.7 | 1494.407 |
| S2020_S_9 | 65151 | 25408700. | 0.531 | 171.0 | 46.4 | 161.4 | 15.0 | 1512.384 |
| S2004_S_52 | 65152 | 26446400. | 0.291 | 292.6 | 333.0 | 165.4 | 204.7 | 1605.886 |
| S2007_S_9 | 65153 | 20174600. | 0.360 | 125.3 | 240.3 | 159.3 | 269.4 | 1069.806 |
| S2004_S_53 | 65154 | 23279800. | 0.240 | 284.3 | 37.3 | 162.6 | 215.7 | 1326.184 |
| S2020_S_10 | 65155 | 25315300. | 0.296 | 213.5 | 158.8 | 165.6 | 243.6 | 1503.968 |
| S2019_S_21 | 65156 | 26439500. | 0.155 | 16.7 | 346.4 | 171.9 | 342.8 | 1605.279 |
| S2006_S_20 | 65157 | 13193700. | 0.206 | 77.4 | 153.1 | 173.1 | 60.9 | 565.7889 |

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mean ecliptic orbital elements
**Epoch:** 2000-01-01.5 TDB
**Ephemeris:** SAT455 [ table legend ]

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) |
|---|---|---|---|---|---|---|---|---|
| S2004_S_54 | 65158 | 11277500. | 0.373 | 108.4 | 69.3 | 48.1 | 80.2 | 447.1410 |
| S2004_S_55 | 65159 | 11294700. | 0.260 | 80.0 | 59.6 | 48.9 | 196.6 | 448.1618 |
| S2004_S_56 | 65160 | 13670200. | 0.339 | 96.4 | 165.7 | 161.6 | 343.2 | 596.6938 |
| S2004_S_57 | 65161 | 18150500. | 0.263 | 60.9 | 202.7 | 167.9 | 39.1 | 913.0736 |
| S2004_S_58 | 65162 | 18254500. | 0.249 | 120.0 | 180.1 | 45.7 | 284.5 | 920.8035 |
| S2004_S_59 | 65163 | 19170700. | 0.262 | 222.1 | 282.6 | 167.3 | 125.2 | 991.1819 |
| S2004_S_60 | 65164 | 19517000. | 0.280 | 264.3 | 357.6 | 173.8 | 321.7 | 1018.128 |
| S2004_S_61 | 65165 | 20986900. | 0.466 | 40.4 | 175.6 | 168.4 | 176.4 | 1135.064 |
| S2005_S_6 | 65166 | 18107300. | 0.084 | 77.4 | 163.4 | 47.7 | 166.5 | 909.5785 |
| S2005_S_7 | 65167 | 18502500. | 0.565 | 132.7 | 181.1 | 34.6 | 269.8 | 939.7451 |
| S2006_S_21 | 65168 | 14976500. | 0.204 | 157.4 | 40.7 | 169.8 | 328.3 | 684.2764 |
| S2006_S_22 | 65169 | 15109500. | 0.246 | 71.7 | 254.1 | 172.0 | 231.1 | 693.4146 |

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) |
|---|---|---|---|---|---|---|---|---|
| S2006_S_23 | 65170 | 18269700. | 0.190 | 72.5 | 309.9 | 43.8 | 257.1 | 921.8632 |
| S2006_S_24 | 65171 | 18210700. | 0.352 | 39.8 | 10.3 | 165.9 | 204.8 | 917.5590 |
| S2006_S_25 | 65172 | 18572400. | 0.303 | 130.1 | 348.5 | 158.8 | 277.4 | 945.0681 |
| S2006_S_26 | 65173 | 18619300. | 0.248 | 2.5 | 201.0 | 171.9 | 147.8 | 948.6722 |
| S2006_S_27 | 65174 | 19205700. | 0.140 | 91.6 | 6.1 | 170.5 | 3.7 | 993.7889 |
| S2006_S_28 | 65175 | 21955100. | 0.210 | 242.2 | 98.4 | 172.9 | 152.8 | 1214.525 |
| S2006_S_29 | 65176 | 25212100. | 0.239 | 301.4 | 101.5 | 156.2 | 82.4 | 1494.783 |
| S2007_S_10 | 65177 | 11364900. | 0.367 | 94.2 | 23.6 | 45.8 | 287.1 | 452.3576 |
| S2007_S_11 | 65178 | 17434400. | 0.499 | 348.9 | 233.9 | 35.5 | 53.9 | 859.5264 |
| S2019_S_22 | 65179 | 11305100. | 0.369 | 100.5 | 223.0 | 47.3 | 21.1 | 448.7750 |
| S2019_S_23 | 65180 | 11310200. | 0.255 | 81.5 | 1.0 | 48.7 | 356.3 | 449.0806 |
| S2019_S_24 | 65181 | 11360500. | 0.345 | 74.7 | 79.3 | 46.7 | 205.3 | 452.0743 |
| S2019_S_25 | 65182 | 11329400. | 0.271 | 73.9 | 358.9 | 48.1 | 213.7 | 450.2181 |
| S2019_S_26 | 65183 | 11390900. | 0.365 | 111.3 | 164.1 | 48.1 | 119.7 | 453.8882 |
| S2019_S_27 | 65184 | 16267000. | 0.420 | 97.2 | 152.1 | 162.1 | 115.1 | 774.6333 |
| S2019_S_28 | 65185 | 17496000. | 0.199 | 95.6 | 307.9 | 158.4 | 268.3 | 864.0938 |
| S2019_S_29 | 65186 | 17353900. | 0.441 | 332.7 | 133.8 | 37.7 | 306.9 | 853.6229 |
| S2019_S_30 | 65187 | 17709900. | 0.107 | 5.3 | 187.8 | 168.3 | 356.8 | 879.9708 |
| S2019_S_31 | 65188 | 17739100. | 0.488 | 37.8 | 35.2 | 39.8 | 173.7 | 882.2410 |
| S2019_S_32 | 65189 | 17960500. | 0.276 | 281.4 | 20.1 | 46.2 | 218.1 | 898.7083 |
| S2019_S_33 | 65190 | 18696100. | 0.289 | 46.9 | 245.6 | 170.4 | 186.4 | 954.5340 |
| S2019_S_34 | 65191 | 18446800. | 0.536 | 226.7 | 78.5 | 37.6 | 169.4 | 935.4472 |
| S2019_S_35 | 65192 | 18557800. | 0.577 | 181.1 | 259.0 | 157.3 | 63.3 | 943.9993 |
| S2019_S_36 | 65193 | 19903200. | 0.161 | 278.1 | 145.8 | 166.9 | 353.4 | 1048.418 |
| S2019_S_37 | 65194 | 19996900. | 0.404 | 355.3 | 135.8 | 149.9 | 273.4 | 1055.777 |
| S2019_S_38 | 65195 | 21998400. | 0.399 | 303.7 | 138.3 | 163.0 | 216.6 | 1218.274 |
| S2019_S_39 | 65196 | 23784500. | 0.098 | 327.0 | 92.5 | 174.5 | 273.1 | 1369.643 |
| S2019_S_40 | 65197 | 24087800. | 0.088 | 242.4 | 347.6 | 161.8 | 44.4 | 1395.877 |
| S2019_S_41 | 65198 | 24493600. | 0.257 | 40.0 | 320.7 | 157.1 | 11.4 | 1431.297 |
| S2019_S_42 | 65199 | 24111600. | 0.121 | 84.5 | 351.4 | 163.2 | 318.2 | 1397.975 |
| S2019_S_43 | 65200 | 26664100. | 0.277 | 108.2 | 306.2 | 165.3 | 116.0 | 1625.900 |
| S2019_S_44 | 65201 | 26796900. | 0.512 | 332.7 | 330.6 | 172.6 | 223.0 | 1638.110 |
| S2020_S_11 | 65202 | 11295600. | 0.372 | 110.4 | 95.9 | 48.2 | 94.2 | 448.2111 |
| S2020_S_12 | 65203 | 11314500. | 0.260 | 81.2 | 350.8 | 50.8 | 70.5 | 449.3271 |
| S2020_S_13 | 65204 | 11415600. | 0.373 | 96.4 | 230.4 | 48.0 | 41.6 | 455.3875 |
| S2020_S_14 | 65205 | 16186200. | 0.313 | 54.7 | 209.3 | 161.7 | 355.4 | 768.8625 |

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) |
|---|---|---|---|---|---|---|---|---|
| S2020_S_15 | 65206 | 16729200. | 0.462 | 275.8 | 141.7 | 37.1 | 338.2 | 807.8243 |
| S2020_S_16 | 65207 | 16963400. | 0.405 | 73.1 | 185.2 | 167.3 | 268.6 | 824.9229 |
| S2020_S_17 | 65208 | 17094200. | 0.378 | 287.2 | 359.0 | 148.9 | 210.9 | 834.4472 |
| S2020_S_18 | 65209 | 17777900. | 0.180 | 165.9 | 218.7 | 168.9 | 204.0 | 885.1153 |
| S2020_S_19 | 65210 | 17726700. | 0.159 | 65.5 | 80.4 | 48.1 | 56.1 | 881.0431 |
| S2020_S_20 | 65211 | 17997300. | 0.133 | 230.5 | 330.0 | 169.8 | 92.9 | 901.5250 |
| S2020_S_21 | 65212 | 18862100. | 0.307 | 188.0 | 95.5 | 169.9 | 328.3 | 967.2604 |
| S2020_S_22 | 65213 | 19443000. | 0.059 | 38.4 | 128.8 | 161.3 | 271.4 | 1012.241 |
| S2020_S_23 | 65214 | 19801500. | 0.089 | 61.5 | 65.1 | 165.0 | 153.6 | 1040.375 |
| S2020_S_24 | 65215 | 20618300. | 0.230 | 114.8 | 301.4 | 159.6 | 42.2 | 1105.408 |
| S2020_S_25 | 65216 | 20763700. | 0.316 | 329.0 | 318.1 | 171.8 | 242.2 | 1117.088 |
| S2020_S_26 | 65217 | 21264400. | 0.273 | 143.0 | 141.3 | 163.2 | 252.9 | 1157.766 |
| S2020_S_27 | 65218 | 21802300. | 0.255 | 314.6 | 32.0 | 145.3 | 324.5 | 1202.016 |
| S2020_S_28 | 65219 | 21993700. | 0.474 | 226.7 | 2.2 | 160.1 | 60.9 | 1217.810 |
| S2020_S_29 | 65220 | 22301400. | 0.047 | 298.7 | 115.9 | 169.1 | 255.9 | 1243.449 |
| S2020_S_30 | 65221 | 21790700. | 0.601 | 352.3 | 60.3 | 154.2 | 132.1 | 1201.018 |
| S2020_S_31 | 65222 | 22457300. | 0.238 | 196.2 | 0.1 | 163.8 | 148.1 | 1256.503 |
| S2020_S_32 | 65223 | 21884100. | 0.502 | 198.2 | 88.9 | 169.1 | 351.6 | 1208.848 |
| S2020_S_33 | 65224 | 22922500. | 0.555 | 4.3 | 165.2 | 162.8 | 161.7 | 1295.850 |
| S2020_S_34 | 65225 | 22435600. | 0.154 | 162.8 | 116.4 | 160.6 | 7.7 | 1254.606 |
| S2020_S_35 | 65226 | 23030300. | 0.225 | 19.8 | 98.0 | 174.9 | 344.7 | 1304.965 |
| S2020_S_36 | 65227 | 22806200. | 0.336 | 139.7 | 216.1 | 168.8 | 26.9 | 1286.029 |
| S2020_S_37 | 65228 | 23751800. | 0.344 | 211.7 | 182.3 | 174.8 | 25.1 | 1366.821 |
| S2020_S_38 | 65229 | 23583900. | 0.513 | 246.3 | 99.4 | 159.7 | 284.0 | 1352.434 |
| S2020_S_39 | 65230 | 24262400. | 0.305 | 28.9 | 165.2 | 160.1 | 318.8 | 1411.145 |
| S2020_S_40 | 65231 | 23785900. | 0.412 | 99.4 | 64.0 | 167.3 | 250.5 | 1369.756 |
| S2020_S_41 | 65232 | 25876400. | 0.402 | 224.9 | 63.1 | 160.2 | 28.3 | 1554.399 |
| S2020_S_42 | 65233 | 25329400. | 0.506 | 59.1 | 278.4 | 157.5 | 15.4 | 1505.311 |
| S2020_S_43 | 65234 | 26657400. | 0.203 | 257.0 | 252.6 | 164.6 | 68.7 | 1625.293 |
| S2020_S_44 | 65235 | 27259400. | 0.199 | 283.8 | 318.0 | 168.5 | 28.0 | 1680.641 |
| S2023_S_1 | 65236 | 11205400. | 0.386 | 296.8 | 263.4 | 48.8 | 159.6 | 442.8632 |
| S2023_S_2 | 65237 | 11309900. | 0.339 | 117.3 | 257.2 | 45.7 | 253.8 | 449.0535 |
| S2023_S_3 | 65238 | 17646400. | 0.178 | 264.0 | 272.6 | 46.9 | 93.7 | 875.0007 |
| S2023_S_4 | 65239 | 17764600. | 0.276 | 9.5 | 123.9 | 170.0 | 264.4 | 884.1111 |
| S2023_S_5 | 65240 | 25583500. | 0.599 | 94.9 | 334.2 | 168.8 | 235.0 | 1528.036 |
| S2023_S_6 | 65241 | 11953100. | 0.336 | 103.7 | 133.4 | 47.4 | 54.4 | 487.9090 |

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) |
|---|---|---|---|---|---|---|---|---|
| S2023_S_7 | 65242 | 12133700. | 0.284 | 102.5 | 102.3 | 44.7 | 289.7 | 499.0097 |
| S2023_S_8 | 65243 | 14018800. | 0.122 | 278.8 | 149.6 | 166.9 | 240.1 | 619.6910 |
| S2023_S_9 | 65244 | 13167500. | 0.141 | 209.8 | 285.2 | 172.2 | 324.8 | 564.1139 |
| S2023_S_10 | 65245 | 15500200. | 0.302 | 124.7 | 70.9 | 163.0 | 308.1 | 720.4931 |
| S2023_S_11 | 65246 | 14046100. | 0.300 | 76.1 | 334.2 | 170.9 | 193.0 | 621.4931 |
| S2023_S_12 | 65247 | 15805900. | 0.601 | 182.5 | 339.7 | 168.8 | 301.0 | 741.9167 |
| S2023_S_13 | 65248 | 15193000. | 0.179 | 107.5 | 173.9 | 168.5 | 134.2 | 699.1799 |
| S2023_S_14 | 65249 | 16853000. | 0.497 | 351.4 | 62.7 | 171.6 | 291.7 | 816.8618 |
| S2023_S_15 | 65250 | 18241300. | 0.549 | 123.2 | 159.7 | 161.9 | 276.6 | 919.9347 |
| S2023_S_16 | 65251 | 17005300. | 0.270 | 116.9 | 80.0 | 162.6 | 270.6 | 827.9111 |
| S2023_S_17 | 65252 | 17385300. | 0.498 | 246.7 | 128.1 | 35.9 | 240.7 | 855.9354 |
| S2023_S_18 | 65253 | 17381700. | 0.448 | 280.0 | 169.7 | 36.7 | 78.1 | 855.6535 |
| S2023_S_19 | 65254 | 17590300. | 0.092 | 205.4 | 325.5 | 48.2 | 50.1 | 870.9229 |
| S2023_S_20 | 65255 | 17261000. | 0.442 | 322.1 | 200.6 | 136.5 | 112.7 | 846.7653 |
| S2023_S_21 | 65256 | 17755400. | 0.077 | 341.9 | 38.7 | 157.3 | 124.2 | 883.3083 |
| S2023_S_22 | 65257 | 18577500. | 0.182 | 266.5 | 4.2 | 47.5 | 121.8 | 945.3701 |
| S2023_S_23 | 65258 | 18783700. | 0.350 | 91.2 | 81.8 | 164.8 | 187.0 | 961.2236 |
| S2023_S_24 | 65259 | 18351800. | 0.374 | 319.7 | 54.4 | 169.7 | 1.4 | 928.2451 |
| S2023_S_25 | 65260 | 19136600. | 0.281 | 293.8 | 186.2 | 166.4 | 216.6 | 988.5278 |
| S2023_S_26 | 65261 | 19894300. | 0.306 | 108.2 | 132.2 | 163.9 | 147.4 | 1047.761 |
| S2023_S_27 | 65262 | 19820100. | 0.652 | 117.0 | 91.2 | 151.1 | 306.6 | 1041.852 |
| S2023_S_28 | 65263 | 19881000. | 0.575 | 2.3 | 189.6 | 168.7 | 59.6 | 1046.555 |
| S2023_S_29 | 65264 | 20042400. | 0.141 | 47.0 | 162.8 | 172.2 | 227.4 | 1059.420 |
| S2023_S_30 | 65265 | 18238300. | 0.493 | 191.3 | 2.7 | 142.4 | 250.0 | 919.7139 |
| S2023_S_31 | 65266 | 20729200. | 0.182 | 240.9 | 308.7 | 163.0 | 261.1 | 1114.288 |
| S2023_S_32 | 65267 | 20454400. | 0.037 | 90.3 | 30.0 | 169.8 | 195.2 | 1092.241 |
| S2023_S_33 | 65268 | 21621900. | 0.665 | 258.6 | 49.1 | 155.8 | 209.3 | 1187.066 |
| S2023_S_34 | 65269 | 20803900. | 0.570 | 275.5 | 293.6 | 168.4 | 303.0 | 1120.383 |
| S2023_S_35 | 65270 | 22269700. | 0.151 | 165.5 | 92.9 | 168.5 | 340.9 | 1240.761 |
| S2023_S_36 | 65271 | 22230600. | 0.359 | 157.0 | 312.2 | 166.3 | 325.1 | 1237.610 |
| S2023_S_37 | 65272 | 19889800. | 0.215 | 241.2 | 244.7 | 172.3 | 63.4 | 1047.390 |
| S2023_S_38 | 65273 | 12823500. | 0.909 | 241.2 | 244.7 | 149.2 | 63.4 | 546.3146 |
| S2023_S_39 | 65274 | 20824500. | 0.124 | 351.9 | 345.1 | 164.8 | 118.3 | 1121.993 |
| S2023_S_40 | 65275 | 21065100. | 0.342 | 334.7 | 3.6 | 169.6 | 271.4 | 1141.481 |
| S2023_S_41 | 65276 | 21286400. | 0.279 | 323.0 | 29.1 | 172.1 | 7.2 | 1159.516 |
| S2023_S_42 | 65277 | 21837000. | 0.059 | 33.2 | 315.9 | 166.7 | 355.1 | 1204.814 |

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) |
|---|---|---|---|---|---|---|---|---|
| S2023_S_43 | 65278 | 22563900. | 0.264 | 218.0 | 92.7 | 170.3 | 247.3 | 1265.566 |
| S2023_S_44 | 65279 | 19292400. | 0.434 | 166.3 | 232.9 | 167.4 | 179.9 | 1000.472 |
| S2023_S_45 | 65280 | 23438400. | 0.633 | 334.1 | 336.8 | 157.4 | 102.2 | 1339.847 |
| S2023_S_46 | 65281 | 24708900. | 0.336 | 100.4 | 230.0 | 143.2 | 306.8 | 1450.271 |
| S2023_S_47 | 65282 | 25102300. | 0.101 | 257.3 | 97.5 | 162.5 | 208.4 | 1485.041 |
| S2023_S_48 | 65283 | 20029200. | 0.022 | 108.2 | 315.6 | 169.7 | 338.7 | 1058.349 |
| S2023_S_49 | 65284 | 21766500. | 0.026 | 110.7 | 82.5 | 171.7 | 193.8 | 1198.989 |
| S2023_S_50 | 65285 | 11656500. | 0.263 | 326.3 | 79.8 | 166.1 | 29.0 | 469.8201 |

mean ecliptic orbital elements
**Epoch:** 2000-01-01.5 TDB
**Ephemeris:** SAT458 [ table legend ]

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) |
|---|---|---|---|---|---|---|---|---|
| S2020_S_45 | 65286 | 23507700. | 0.199 | 195.7 | 190.2 | 172.8 | 183.1 | 1345.791 |
| S2020_S_46 | 65287 | 18892100. | 0.207 | 331.2 | 16.0 | 167.3 | 323.4 | 969.5965 |
| S2020_S_47 | 65288 | 21397400. | 0.564 | 5.4 | 283.9 | 146.1 | 351.6 | 1168.577 |
| S2020_S_48 | 65289 | 11355100. | 0.373 | 114.9 | 21.9 | 45.9 | 253.5 | 451.7528 |
| S2023_S_51 | 65290 | 23431500. | 0.191 | 303.4 | 310.9 | 163.3 | 213.5 | 1339.210 |
| S2023_S_52 | 65291 | 22528000. | 0.124 | 88.7 | 267.3 | 146.2 | 42.8 | 1262.343 |
| S2023_S_53 | 65292 | 17181100. | 0.103 | 55.0 | 179.0 | 171.2 | 314.6 | 840.8181 |
| S2023_S_54 | 65293 | 17485100. | 0.480 | 60.4 | 48.2 | 37.8 | 120.8 | 863.3486 |
| S2023_S_55 | 65294 | 16875100. | 0.491 | 60.8 | 68.8 | 35.9 | 191.1 | 818.5125 |
| S2023_S_56 | 65295 | 11287500. | 0.358 | 89.8 | 341.9 | 45.4 | 271.0 | 447.7458 |
| S2023_S_57 | 65296 | 20536100. | 0.245 | 183.5 | 318.0 | 168.0 | 86.3 | 1098.828 |

## Satellites of Uranus

mean equatorial orbital elements
**Epoch:** 2000-01-01.5 TDB
**Ephemeris:** URA182 [ table legend ]

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) | P ( |
|---|---|---|---|---|---|---|---|---|---|
| Ariel | 701 | 190929. | 0.001 | 9.6 | 193.5 | 0.0 | 0.0 | 2.520379 | 28 |
| Umbriel | 702 | 265986. | 0.004 | 183.4 | 253.0 | 0.1 | 174.8 | 4.144177 | 64 |

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) | P ( |
|---|---|---|---|---|---|---|---|---|---|
| Titania | 703 | 436298. | 0.002 | 184.0 | 68.1 | 0.1 | 29.5 | 8.705869 | 57 |
| Oberon | 704 | 583511. | 0.002 | 132.2 | 143.6 | 0.1 | 76.8 | 13.463237 | 15 |
| Miranda | 705 | 129846. | 0.001 | 154.8 | 73.0 | 4.4 | 100.9 | 1.413479 | 8 |
| Puck | 715 | 86004. | 0.000 | 0.0 | 50.1 | 0.3 | 216.1 | 0.761833 | 2 |

mean orbital elements referred to the local Laplace (/glossary/lp.html) planes
**Epoch:** 2025-01-01.0 TDB
**Ephemeris:** URA184 [ table legend ]

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) | $P_{aps}$ (yr) |
|---|---|---|---|---|---|---|---|---|---|
| Cordelia | 706 | 49755. | 0.000 | 0.0 | 287.4 | 0.2 | 1.1 | 0.3347 | 0.0 |
| Ophelia | 707 | 53765. | 0.011 | 19.3 | 213.4 | 0.2 | 151.6 | 0.3764 | 0.4 |
| Bianca | 708 | 59170. | 0.006 | 328.0 | 109.1 | 2.3 | 272.5 | 0.4347 | 0.5 |
| Cressida | 709 | 61770. | 0.004 | 87.9 | 0.5 | 1.8 | 308.2 | 0.4639 | 0.6 |
| Desdemona | 710 | 62663. | 0.007 | 137.0 | 230.0 | 3.1 | 283.9 | 0.4736 | 0.6 |
| Juliet | 711 | 64362. | 0.006 | 274.9 | 319.8 | 3.0 | 141.0 | 0.4931 | 0.7 |
| Portia | 712 | 66101. | 0.004 | 31.6 | 310.1 | 2.7 | 146.7 | 0.5132 | 0.8 |
| Rosalind | 713 | 69930. | 0.003 | 231.8 | 287.7 | 1.7 | 330.0 | 0.5583 | 0.9 |
| Belinda | 714 | 75258. | 0.002 | 58.6 | 226.4 | 1.4 | 96.2 | 0.6236 | 1.2 |
| Puck | 715 | 86007. | 0.009 | 337.0 | 264.1 | 1.1 | 111.2 | 0.7618 | 1.9 |
| Perdita | 725 | 76418. | 0.005 | 296.9 | 168.0 | 1.6 | 270.4 | 0.6382 | 1.3 |
| Mab | 726 | 97737. | 0.006 | 318.6 | 250.8 | 1.8 | 307.8 | 0.9229 | 3.1 |
| Cupid | 727 | 74396. | 0.007 | 168.0 | 3.3 | 2.0 | 31.1 | 0.6125 | 1.2 |
| S2025_U_1 | 75052 | 57844. | 0.039 | 313.9 | 275.6 | 4.0 | 70.8 | 0.4201 | 0.5 |

mean ecliptic orbital elements
**Epoch:** 2020-01-01.0 TDB
**Ephemeris:** URA117 [ table legend ]

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) | $P_a$ ( |
|---|---|---|---|---|---|---|---|---|---|
| Caliban | 716 | 7167000. | 0.200 | 349.7 | 241.2 | 141.4 | 174.9 | 580. | 87 |
| Sycorax | 717 | 12193200. | 0.520 | 25.4 | 332.1 | 157.0 | 267.1 | 1286. | 13 |
| Prospero | 718 | 16221000. | 0.441 | 180.4 | 197.6 | 149.4 | 324.5 | 1974. | 1 |
| Setebos | 719 | 17519800. | 0.579 | 356.1 | 148.0 | 153.9 | 244.7 | 2215. | 4 |

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) | $P_a$ ( |
|---|---|---|---|---|---|---|---|---|---|
| Stephano | 720 | 7951400. | 0.235 | 17.1 | 164.4 | 143.6 | 193.3 | 677. | 54 |
| Trinculo | 721 | 8502600. | 0.220 | 162.2 | 55.6 | 167.1 | 196.5 | 749. | 25 |
| Francisco | 722 | 4275700. | 0.144 | 137.6 | 288.4 | 146.8 | 101.9 | 267. | 12 |
| Margaret | 723 | 14425000. | 0.642 | 90.7 | 115.9 | 60.5 | 0.9 | 1655. | |
| Ferdinand | 724 | 20421400. | 0.395 | 166.9 | 172.3 | 169.2 | 223.9 | 2788. | 7 |
| S2023_U1 | 75051 | 7976600. | 0.250 | 158.7 | 101.8 | 143.9 | 260.2 | 681. | 50 |

## Satellites of Neptune

mean orbital elements referred to the local Laplace (/glossary/lp.html) planes
**Epoch:** 2000-01-01.5 TDB
**Ephemeris:** NEP097 [ table legend ]

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) | $P_a$ ( |
|---|---|---|---|---|---|---|---|---|---|
| Triton | 801 | 354800. | 0.000 | 0.0 | 63.0 | 157.3 | 178.1 | 5.876994 | 0. |
| Naiad | 803 | 48200. | 0.000 | 0.0 | 89.7 | 4.7 | 41.4 | 0.293980 | 0. |
| Thalassa | 804 | 50100. | 0.000 | 0.0 | 165.7 | 0.2 | 130.6 | 0.311078 | 0. |
| Despina | 805 | 52500. | 0.000 | 0.0 | 125.1 | 0.0 | 0.0 | 0.334656 | 0. |
| Galatea | 806 | 62000. | 0.000 | 0.0 | 86.7 | 0.0 | 0.0 | 0.428744 | 0. |
| Larissa | 807 | 73500. | 0.001 | 247.3 | 165.5 | 0.2 | 312.0 | 0.554989 | 1. |
| Proteus | 808 | 117600. | 0.000 | 0.0 | 276.8 | 0.0 | 0.0 | 1.122315 | 0. |
| Hippocamp | 814 | 105300. | 0.001 | 346.4 | 286.5 | 0.3 | 0.5 | 0.950390 | 8. |

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mean ecliptic orbital elements
**Epoch:** 2020-01-01.0 TDB
**Ephemeris:** NEP105 [ table legend ]

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) |
|---|---|---|---|---|---|---|---|---|
| Nereid | 802 | 5513900. | 0.751 | 296.8 | 318.5 | 5.1 | 319.5 | 360.133039 |

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mean ecliptic orbital elements
**Epoch:** 2020-01-01.0 TDB
**Ephemeris:** NEP104 [ table legend ]

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) | P ( |
|---|---|---|---|---|---|---|---|---|---|
| Halimede | 809 | 16590500. | 0.521 | 157.9 | 135.8 | 119.6 | 198.5 | 1879. | 5 |
| Psamathe | 810 | 47646600. | 0.413 | 148.2 | 183.2 | 127.8 | 302.9 | 9149. | 1 |
| Sao | 811 | 22239900. | 0.296 | 93.5 | 178.5 | 50.2 | 41.8 | 2919. | |
| Laomedeia | 812 | 23499900. | 0.419 | 146.2 | 248.1 | 36.9 | 57.6 | 3168. | 2 |
| Neso | 813 | 49897800. | 0.455 | 89.8 | 13.8 | 128.4 | 55.4 | 9805. | |
| S2002_N5 | 85051 | 23414700. | 0.433 | 59.1 | 303.2 | 46.3 | 258.3 | 3151. | 2 |
| S2021_N1 | 85052 | 50700200. | 0.503 | 90.4 | 237.1 | 135.2 | 258.9 | 10043. | |

## Satellites of Pluto

mean equatorial orbital elements
**Epoch:** 2000-01-01.5 TDB
**Ephemeris:** PLU060 [ table legend ]

| Satellite | Code | a (km) | e | ω (deg) | M (deg) | i (deg) | node (deg) | P (days) | $P_{apsis}$ (yr) |
|---|---|---|---|---|---|---|---|---|---|
| Charon | 901 | 19600. | 0.000 | 0.0 | 304.1 | 0.0 | 0.0 | 6.387222 | - |
| Styx | 905 | 43200. | 0.025 | 322.5 | 358.1 | 0.0 | 0.0 | 20.16 | - |
| Nix | 902 | 49300. | 0.015 | 31.4 | 338.2 | 0.0 | 0.0 | 24.85 | - |
| Kerberos | 904 | 58300. | 0.010 | 32.1 | 276.1 | 0.4 | 314.3 | 32.17 | - |
| Hydra | 903 | 65200. | 0.009 | 139.3 | 335.0 | 0.3 | 114.3 | 38.20 | - |

## Table Column Descriptions

| | |
|---|---|
| **Satellite** | IAU name or designation of the satellite |
| **Code** | JPL satellite ID |
| **Ephemeris** | planetary satellite ephemeris ID used to derive the mean elements |
| **Frame** | reference frame of the mean elements |
| **Epoch** | epoch of the osculating mean elements |
| **a** | semi-major Axis |
| **e** | eccentricity |
| **ω** | argument of periapsis |
| **M** | mean anomaly |
| **i** | inclination with respect to the reference plane |
| **node** | longitude of the ascending node measured from the node of the reference plane on the ICRF equator |
| **P** | sidereal period |
| **$P_{apsis}$** | argument of periapsis precession period |

| | |
|---|---|
| **$P_{node}$** | longitude of the ascending node precession period |
| **R.A.** | right ascension of the Laplace-plane pole postion w.r.t. ICRF |
| **Dec.** | declination of the Laplace-plane pole postion w.r.t. ICRF |
| **Tilt** | angle between the planet's equatorial and Laplace planes |

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